A13483W1

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B4. SUB-ATOMIC PHYSICS

TRINITY TERM 2016

Wednesday, 8 June, 2.30 pm – 4.30 pm

10 minutes reading time

Answer **two** *questions.*

Start the answer to each question in a **fresh book.**

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

1. Give the quark composition of the neutron, its spin, and the number of hadrons in the multiplet in which it sits. [3]

A naïve order-of-magnitude estimate of the neutron mass can be obtained by assuming the constituent quarks are free particles, uniformly distributed in a sphere of radius 1.5 fm:

(a) Calculate the root-mean-square of the radial position of a constituent quark. Take this to be the uncertainty on its radial position.

(b) Estimate the uncertainty on the quark's radial momentum. Assume this is equal to the mean momentum (valid for an exponential momentum distribution) and calculate the energy of the quark, taking its mass to be 5 MeV/c^2.

(c) Sum over the constituent quark energies to obtain an estimate of the neutron mass. What is the ratio of the true neutron mass (940 MeV/c^2) to the estimated mass? What additional information could help improve the estimate? [7]

The free neutron is unstable. Draw a diagram showing the constituent quarks and the decay through an intermediate vector boson. Why is there not a similar process producing proton decay? [4]

Consider the kinematics of neutron decay. What kinematic configurations correspond to the minimum and maximum total energy of the electron in the neutron's rest frame? For these configurations, state which particles are at rest and give the relationship between the momenta of particles not at rest. Using the conservation of four-momentum, calculate the minimum and maximum energies of the electron in terms of the rest masses of the electron (m_e), neutron (m_n), and proton (m_p). Neglect the very small (though non-zero) neutrino mass. [9]

The free neutron lifetime is $\approx$ 15 minutes. Convert this to a decay width in units of electron-volts. [2]

2. The lightest mesons are important particles, binding the nucleus together via the strong interaction and demonstrating features of the weak force in their decay. Draw a diagram for the meson nonet containing kaons and pions. Mark the lines of constant charge and strangeness, showing their values. [4]

Draw a Feynman diagram for each of the decays $\pi^+ \to \mu^+\nu_\mu$, $K^+ \to \mu^+\nu_\mu$, and $K^+ \to \pi^+\pi^0$. Consider the Cabbibo and phase space factors in the corresponding lifetime calculations. What factor suppresses the pion decay relative to the kaon decay to $\mu\nu$, and what factor suppresses the kaon decays relative to the pion decay? [7]

Consider an experimental design where neutral kaons are produced via $\pi^+\pi^-$ scattering.

(a) Each pion beam has an energy equal to exactly half the kaon mass times c^2; take this energy to be 249 MeV. The beams are made to collide with momenta in opposite directions. Using the formula for the resonant production of a broad relativistic resonance,

$$\sigma(E) \propto (2J+1)\frac{B_{in}B_{out}\Gamma_{tot}^2}{(E^2-m^2)^2+E^2\Gamma_{tot}^2},$$

determine the ratio of ρ^0 to K^0 meson production cross sections. The angular momentum J is 1 for the ρ^0 meson and 0 for the K^0 meson. The branching ratios are $B(\rho^0 \to \pi^+\pi^-) = 100\%$ and $B(K^0 \to \pi^+\pi^-) = 69\%$. Take $m_\rho = 769$ MeV$/c^2$, $\Gamma_\rho = 147$ MeV, and $\Gamma_K = 7.4 \times 10^{-12}$ MeV. [5]

(b) Calculate the ratio of ρ^0 to K^0 production cross sections if the beam energy is increased to 250 MeV. Why is it impractical to study resonant K^0 production in $\pi^+\pi^-$ scattering? [5]

(c) The pions are produced by bombarding a carbon target with protons; they are approximately collinear with the proton beam. Pions with a total energy of 249 MeV are swept away by a perpendicular magnetic field of strength 1 Tesla, giving $qB = 1$ eV s m^{-2}. Calculate the radius of curvature of the pion trajectories. [4]

3. Fusion and fission are important reactions, with practical applications.

(a) Write down the initial and final states of the proton-proton fusion chain reaction that produces deuterium in the Sun. Give the energy Q released in the reaction. [2]

(b) A chain of reactions produces helium in the Sun. The initial and final states of this chain are:

$$4(^1_1\mathrm{H}) \rightarrow ^4_2\mathrm{He} + 2e^+ + 2\nu_e + 2\gamma\ (Q = 24.68\ \mathrm{MeV}).$$

The positrons annihilate with electrons in the plasma to produce 1.02 MeV of radiation per positron. Each neutrino carries 0.26 MeV of energy; the neutrinos escape into space and are not measured. The measured rate of energy produced by the Sun is 3.86×10^{26} watts (i.e. including the energy from electron-positron annihilation but excluding neutrino radiation). Assuming the energy produced in other reactions is negligible, and the Sun has been burning for 4.6 billion years, calculate the number of hydrogen nuclei burned by the Sun to date. If the Sun initially had 9×10^{56} hydrogen nuclei, how much longer could it produce helium via this reaction if all the hydrogen were burned? [6]

(c) The semi-empirical mass formula is

$$\begin{aligned} M(Z,A) &= Z(m_p + m_e) + (A - Z)m_n - a_v A + a_s A^{2/3} + \\ & \quad a_c \frac{Z^2}{A^{1/3}} + a_a \frac{(Z - A/2)^2}{A} \pm a_p A^{-1/2}, \end{aligned}$$

where $a_v = 15.56$ MeV, $a_s = 17.23$ MeV, $a_c = 0.697$ MeV, $a_a = 93.14$ MeV, and $a_p = 12$ MeV. State the source of each of the seven terms in the equation. Give a physical explanation for the terms with a_v, a_s, a_c, and a_a (including their signs). [6]

(d) Consider neutron-induced fission of $^{235}_{92}\mathrm{U}$ with products $^{92}_{37}\mathrm{Rb}$, $^{140}_{55}\mathrm{Cs}$, and four neutrons. Calculate the energy released per fission, in units of MeV. If a nuclear reactor produces 100 MW of power from the energy produced by this fission, calculate the total number of uranium atoms burned by the reactor each year. Convert this to grams. [7]

(e) In a nuclear reactor the neutrons from the fission of $^{235}_{92}\mathrm{U}$ have a kinetic energy of 2 MeV on average, but need to be thermalized to ≈ 0.025 eV in order to induce further $^{235}_{92}\mathrm{U}$ fissions at an appreciable rate. A graphite moderator is used to thermalize the neutrons. The average kinetic energy of the neutron after the collision is

$$E_f = \frac{m_C^2 + m_n^2}{(m_C + m_n)^2} E_i,$$

where the neutron mass is $m_n = 0.94\ \mathrm{GeV}/c^2$ and the carbon mass is $m_C = 11.18\ \mathrm{GeV}/c^2$. Determine the number of elastic collisions required to thermalize a given neutron. [4]

4. Measurements of the W and Z boson widths can constrain the number of generations of low-mass particles.

(a) List the Z and W boson decays to the first generation of leptons and quarks. Calculate the branching ratio for the W boson decay to a lepton pair, and for it to decay to a quark pair. [5]

(b) The partial width of a W boson decaying to an electron and a neutrino is 232 MeV. The total measured width of the W boson is 2085 MeV. What does the measurement imply about the number of distinguishable fermion pairs contributing to the W boson width? [3]

(c) The observed W and Z boson decay widths are consistent with decays to three generations of quarks and leptons. Consider a fourth generation of quarks, T and B, charged lepton σ, and neutrino N, with the same quantum numbers as the other generations but different masses. Determine the constraints on the fourth generation masses m_T, m_B, m_σ, and m_N, implied by the W and Z boson decay width observations. [4]

(d) The W boson total decay width can be indirectly determined by measuring the ratio of production of $W \to l\nu$ to $Z \to ll$ in $p\bar{p}$ collisions, using the relativistic Breit-Wigner distribution,

$$\sigma(E) \propto (2J+1)\frac{B_{in}B_{out}\Gamma_{tot}^2}{(E^2-m^2)^2+m^2\Gamma_{tot}^2},$$

together with the standard model calculations of the partial widths, $\Gamma_{in}^{W,Z}$ and $\Gamma_{out}^{W,Z}$, the precisely measured Z boson mass and total width, m_Z and Γ_Z, and the W boson mass, m_W. Find an expression for the W boson width (Γ_W) in terms of these quantities. Assume that each cross section measurement is dominated by collisions at the center of mass energy equal to the mass of the produced boson, and that the proportionality factor (which depends on parton distribution functions that can differ for W and Z boson production) is a known input whose effect can be neglected. [4]

(e) The ratio of production of $W \to l\nu$ to $Z \to ll$ is also sensitive to the presence of a fourth generation, and is independent of the sensitivity from direct width measurements. Calculate the expected W and Z boson widths in the case where the leptons from a fourth generation contribute to the W and Z boson widths. Assume all decays are kinematically allowed and take the partial widths to be equal to those of the first generation [use $\Gamma_Z = 2495$ MeV, $\Gamma_{Z\to\nu\nu} = 501$ MeV, and $\Gamma_{Z\to ee} = 84$ MeV]. What is the fractional effect of such a fourth generation on the ratio of $W \to l\nu$ to $Z \to ll$ cross sections? [5]

(f) The CDF experiment at the $p\bar{p}$ Tevatron collider measured the ratio of $W \to l\nu$ to $Z \to ll$ cross sections with a standard deviation of 1.9%. LEP directly measured the Z boson width with a standard deviation of 2.3 MeV. Calculate the expected number of standard deviations from the standard model prediction that would have been observed in each of these measurements if there were a fourth generation of leptons contributing to the W and Z boson widths. [4]